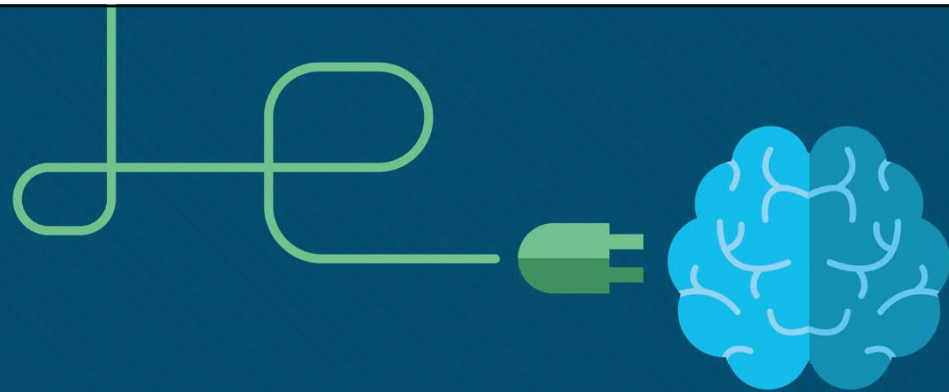




Module 1: Networking Today

Introduction to Networks v7.0
(ITN)



Module Objectives

Module Title: Networking Today

Module Objective: Explain the advances in modern technologies.

Topic Title	Topic Objective
Networks Affect our Lives	Explain how networks affect our daily lives.
Network Components	Explain how host and network devices are used.
Network Representations and Topologies	Explain network representations and how they are used in network topologies.
Common Types of Networks	Compare the characteristics of common types of networks.
Internet Connections	Explain how LANs and WANs interconnect to the internet.
Reliable Networks	Describe the four basic requirements of a reliable network.
Network Trends	Explain how trends such as BYOD, online collaboration, video, and cloud computing are changing the way we interact.
Network Security	Identify some basic security threats and solution for all networks.
The IT Professional	Explain employment opportunities in the networking field.

1.1 Networks Affect Our Lives

Networking Today

Networks Connect Us

- Communication is almost as important to us as our reliance on air, water, food, and shelter.
- In today's world, through the use of networks, we are connected like never before.

Networking Today

No Boundaries

- Advancements in networking technologies are helping create a world without boundaries.
- The immediate nature of communications over the Internet encourages global communities.
- Cisco refers to the impact of the Internet and networks on people the “human network”.



1.2 Network Components

Network Components

Host Roles

Every **computer** on a network is called a **host** or **end device**.

Servers are computers that **provide information to end devices**:

- email servers
- web servers
- file server

Clients are computers that send requests to the servers to retrieve information:

- web page from a web server
- email from an email server

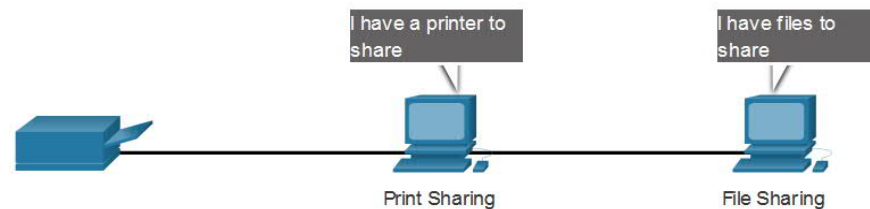


Server Type	Description
Email	Email server runs email server software. Clients use client software to access email.
Web	Web server runs web server software. Clients use browser software to access web pages.
File	File server stores corporate and user files. The client devices access these files.

Network Components

Peer-to-Peer

It is possible to have a **device be a client and a server** in a Peer-to-Peer Network. This type of network design is only recommended for very small networks.

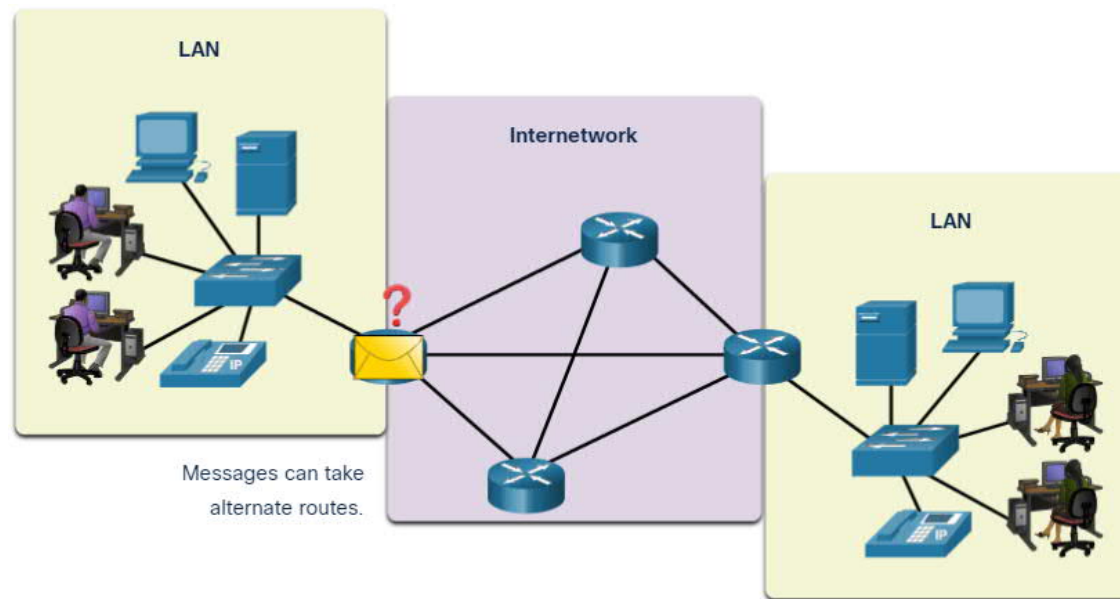


Advantages	Disadvantages
Easy to set up	No centralized administration
Less complex	Not as secure
Lower cost	Not scalable
Used for simple tasks: transferring files and sharing printers	Slower performance

Network Components

End Devices

An end device is where a message originates from or where it is received. Data originates with an end device, flows through the network, and arrives at an end device.



Network Components

Intermediary Network Devices

An intermediary device interconnects end devices. Examples include switches, wireless access points, routers, and firewalls.

Management of data as it flows through a network is also the role of an intermediary device, including:

- Regenerate and retransmit data signals.
- Maintain information about what pathways exist in the network.
- Notify other devices of errors and communication failures.

Intermediary
Devices



Wireless Router



LAN Switch



Router



Multilayer Switch



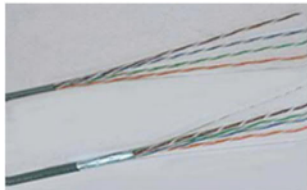
Firewall Appliance

Network Components

Network Media

Communication across a network is carried through a medium which allows a message to travel from source to destination.

Media Types	Description
Metal wires within cables	Uses electrical impulses
Glass or plastic fibers within cables (fiber-optic cable)	Uses pulses of light.
Wireless transmission	Uses modulation of specific frequencies of electromagnetic waves.



1.3 Network Representations and Topologies

Network Representations and Topologies

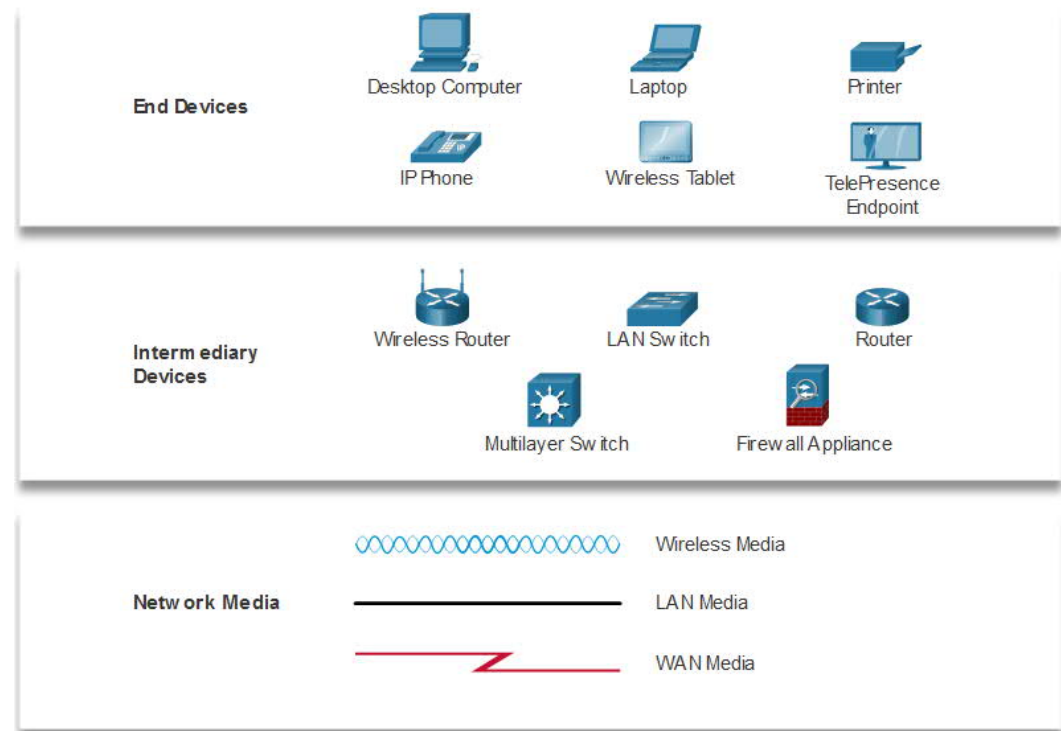
Network Representations

Network diagrams, often called topology diagrams, use symbols to represent devices within the network.

Important terms to know include:

- Network Interface Card (NIC)
- Physical Port
- Interface

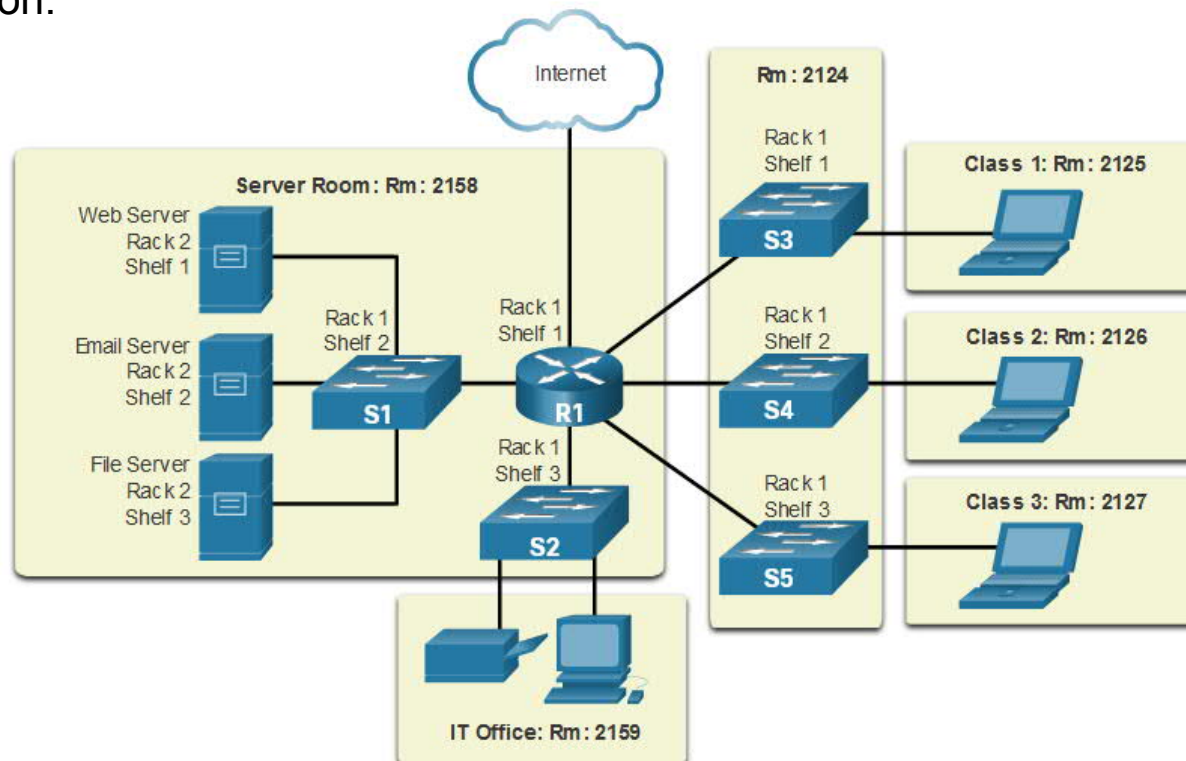
Note: Often, the terms port and interface are used interchangeably



Network Representations and Topologies

Topology Diagrams

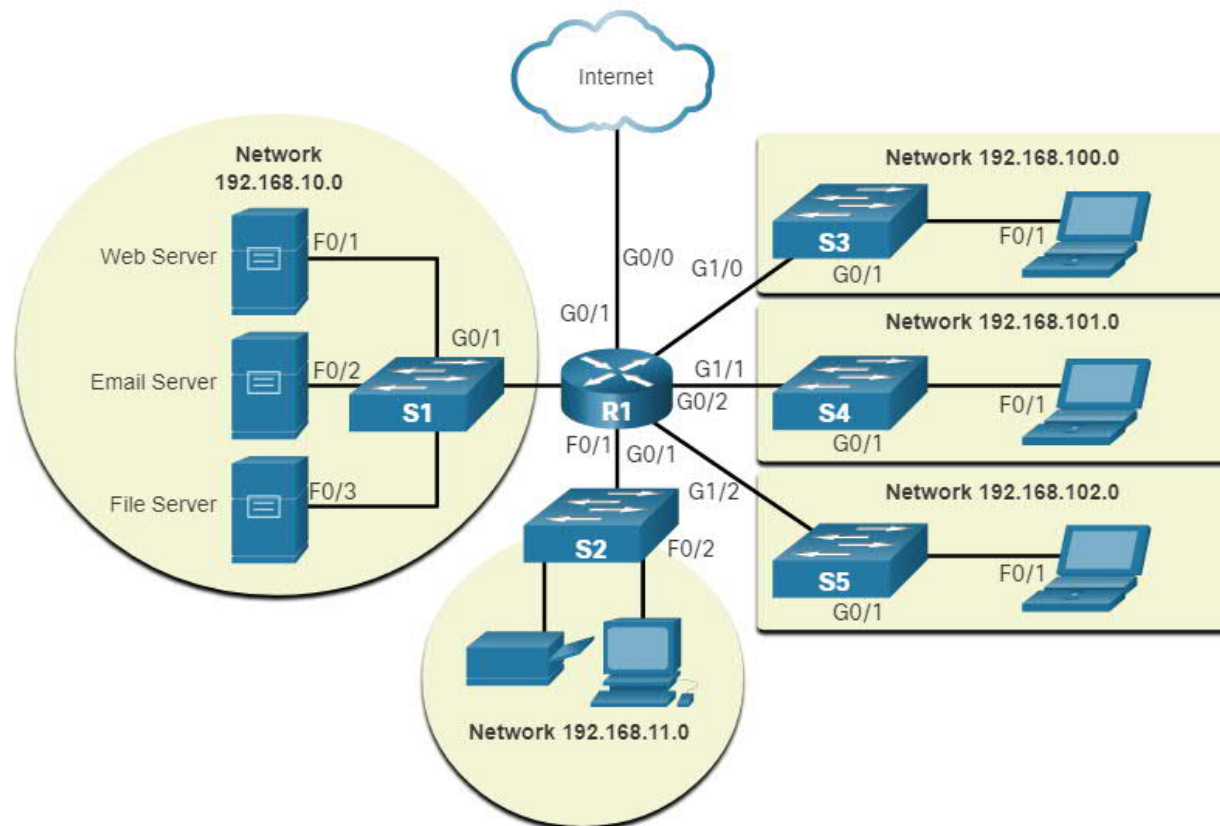
Physical topology diagrams illustrate the **physical location** of intermediary devices and cable installation.



Network Representations and Topologies

Topology Diagrams

Logical topology diagrams illustrate **devices, ports, and the addressing scheme** of the network.



1.4 Common Types of Networks

Common Types of Networks

Networks of Many Sizes



Small Home



SOHO



Medium/Large



World Wide

- **Small Home Networks** – connect a few computers to each other and the Internet
- **Small Office/Home Office** – enables computer within a home or remote office to connect to a corporate network
- **Medium to Large Networks** – many locations with hundreds or thousands of interconnected computers
- **World Wide Networks** – connects hundreds of millions of computers world-wide – such as the internet

Common Types of Networks

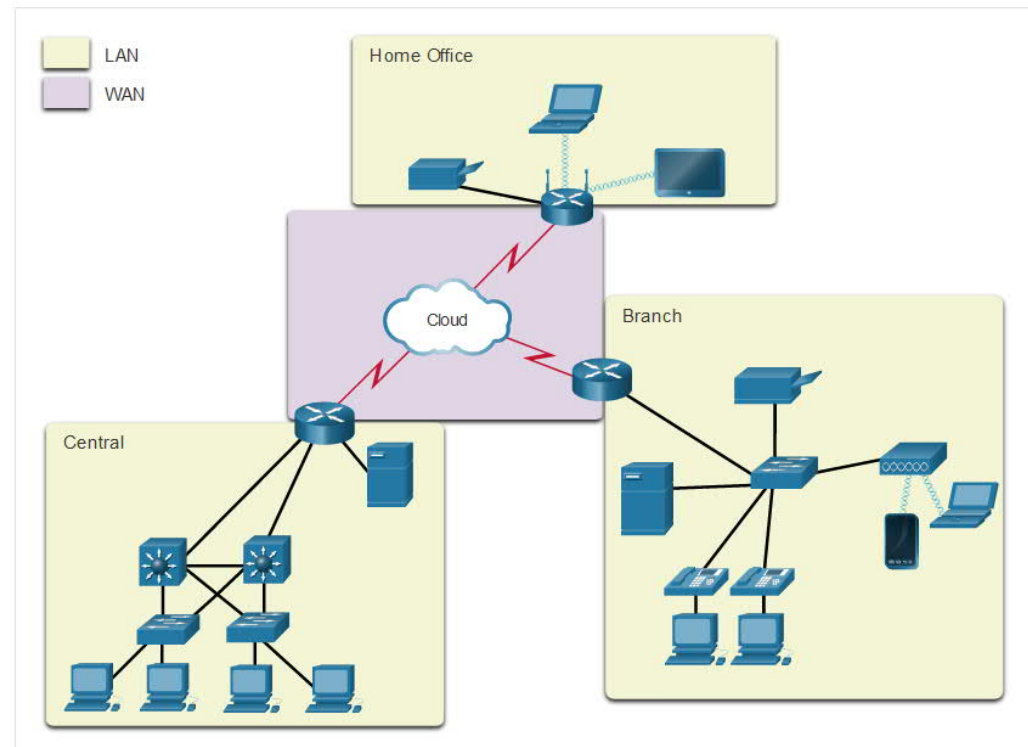
LANs and WANs

Network infrastructures vary greatly in terms of:

- Size of the area covered
- Number of users connected
- Number and types of services available
- Area of responsibility

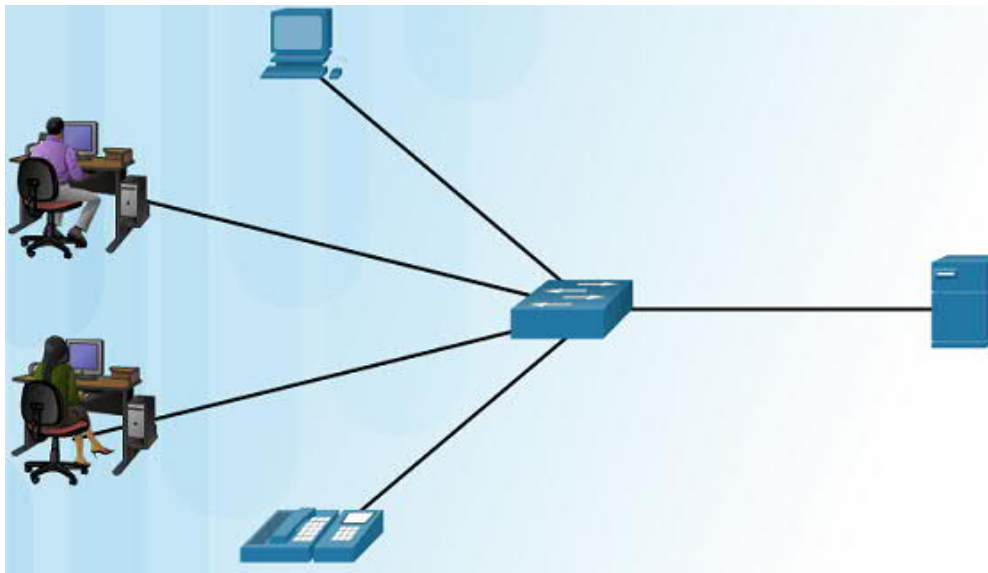
Two most common types of networks:

- Local Area Network (**LAN**)
- Wide Area Network (**WAN**).



Common Types of Networks

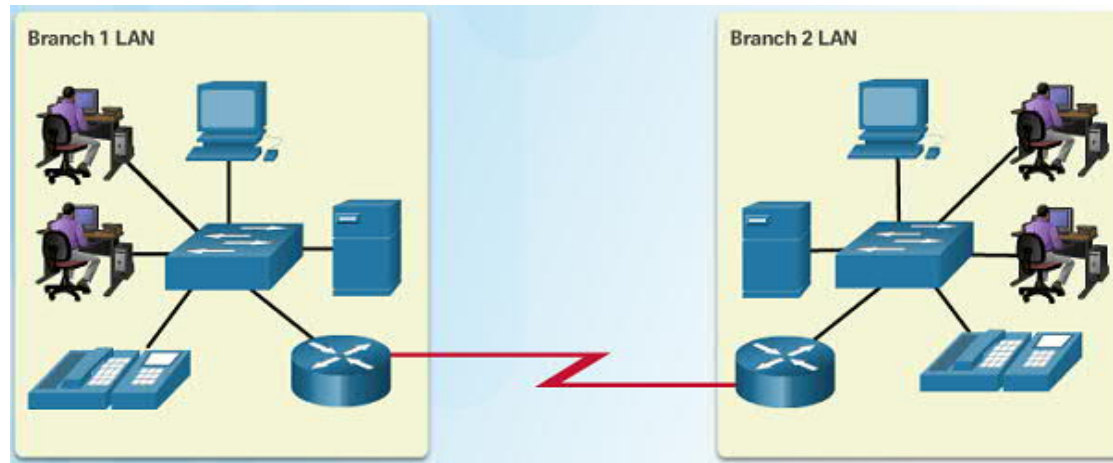
Local Area Networks



- Three characteristics of LANs:
 - Spans a **small geographic area** such as a home, school, office building, or campus.
 - Usually **administered by a single organization** or individual.
 - Provides **high speed bandwidth** to end and intermediary devices within the network.

LANs and WANs

Wide Area Networks



- Three characteristics of WANs:
 - WANs **interconnect LANs** over wide geographical areas such as between cities, states, or countries.
 - Usually **administered by multiple service providers**.
 - WANs typically provide **slower speed links** between LANs.

Common Types of Networks

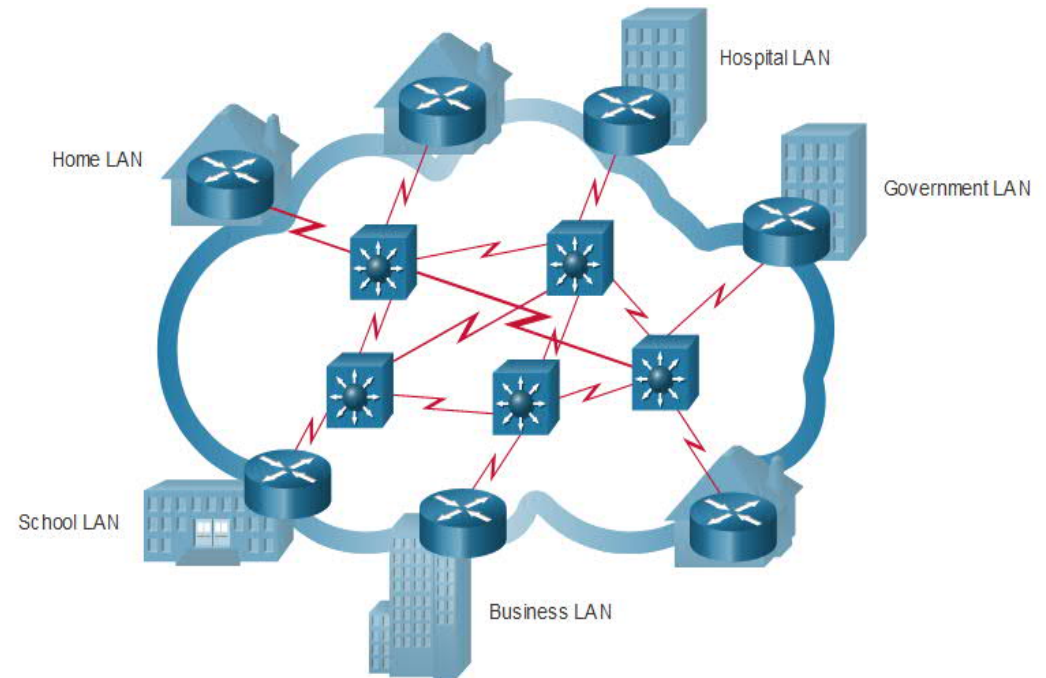
The Internet

The internet is a **worldwide collection of interconnected LANs and WANs**.

- LANs are connected to each other using WANs.
- WANs may use copper wires, fiber optic cables, and wireless transmissions.

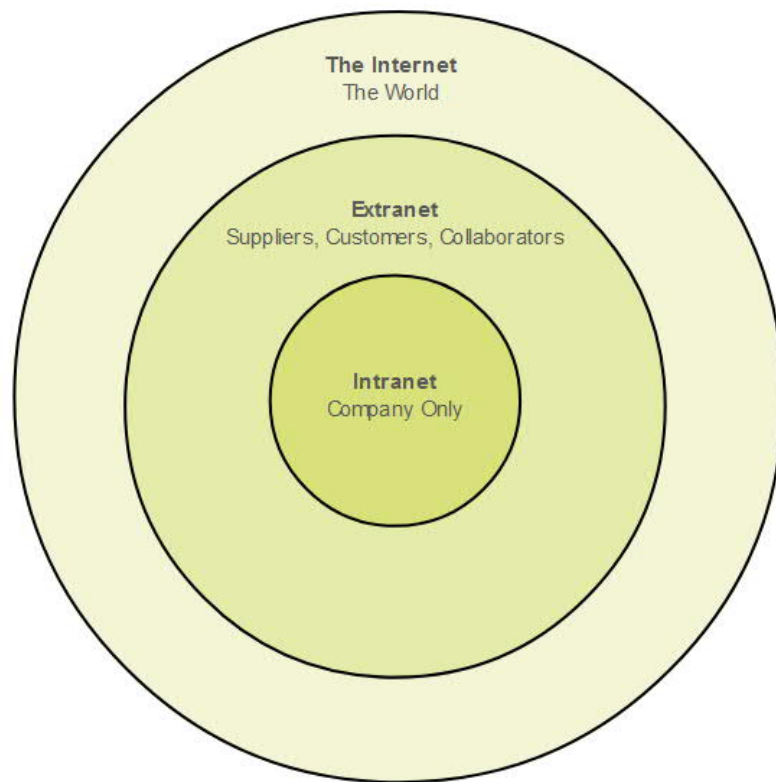
The internet is not owned by any individual or group. The following groups were developed to help maintain structure on the internet:

- IETF (Internet Engineering Task Force)
- ICANN (Internet Corporation for Assigned Names and Numbers)
- IAB (Internet Architecture Board)



Common Types of Networks

Intranets and Extranets



- An **intranet** is a private collection of LANs and WANs internal to an organization that is meant to be accessible only to the organizations members or others with authorization.
- An organization might use an **extranet** to provide secure access to their network for individuals who work for a different organization that need access to their data on their network.

1.5 Internet Connections

Internet Connections

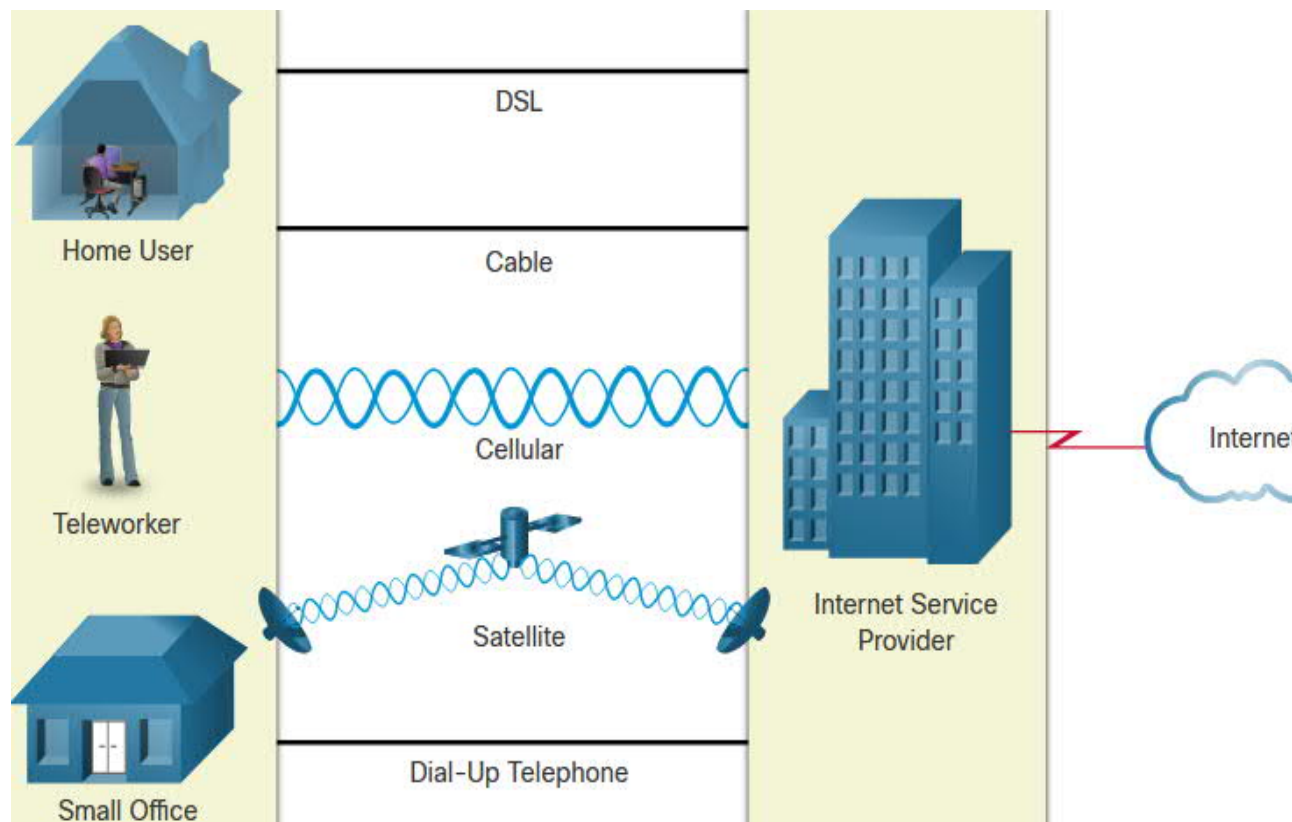
Internet Access Technologies



There are many ways to connect users and organizations to the internet:

- Popular services for home users and small offices include **broadband cable, broadband digital subscriber line (DSL), wireless WANs, and mobile services.**
- Organizations need faster connections to support IP phones, video conferencing and data center storage.
- Business-class interconnections are usually provided by service providers (SP) and may include: business DSL, leased lines, and Metro Ethernet.

Home and Small Office Internet Connections



Home and Small Office Internet Connections

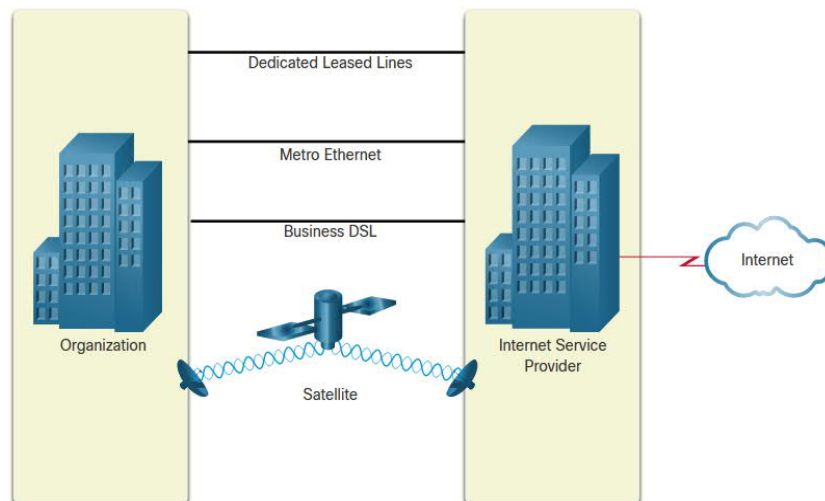
Connection	Description
Cable	high bandwidth, always on, internet offered by cable television service providers.
DSL	high bandwidth, always on, internet connection that runs over a telephone line.
Cellular	uses a cell phone network to connect to the internet.
Satellite	major benefit to rural areas without Internet Service Providers.
Dial-up telephone	an inexpensive, low bandwidth option using a modem.

Internet Connections

Businesses Internet Connections

Corporate business connections may require:

- higher bandwidth
- dedicated connections
- managed services

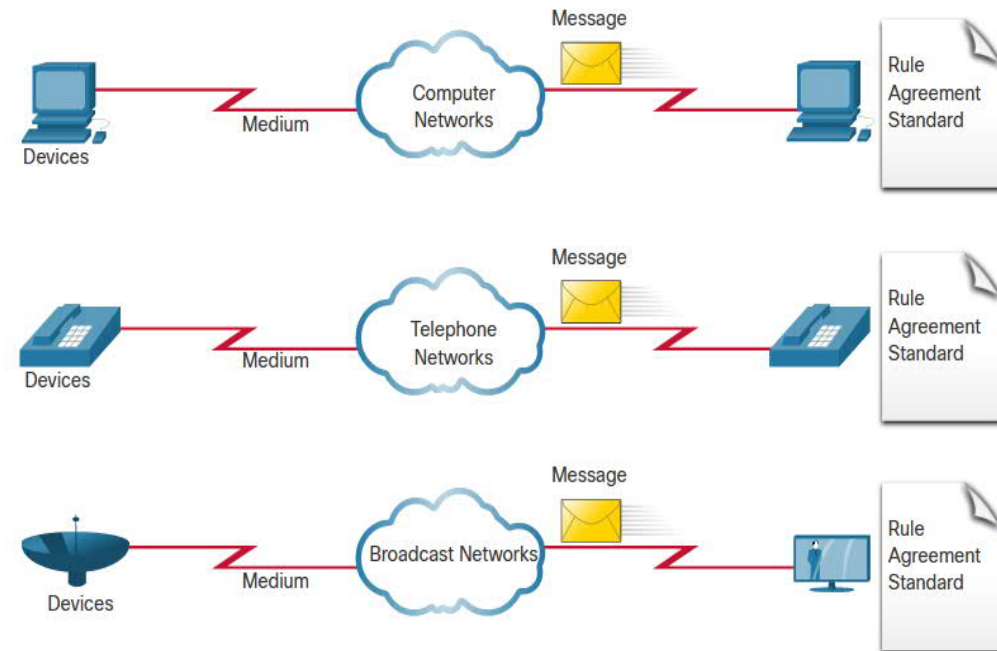


Type of Connection	Description
Dedicated Leased Line	These are reserved circuits within the service provider's network that connect distant offices with private voice and/or data networking.
Ethernet WAN	This extends LAN access technology into the WAN.
DSL	Business DSL is available in various formats including Symmetric Digital Subscriber Lines (SDSL).
Satellite	This can provide a connection when a wired solution is not available.

Internet Connections

The Converging Network

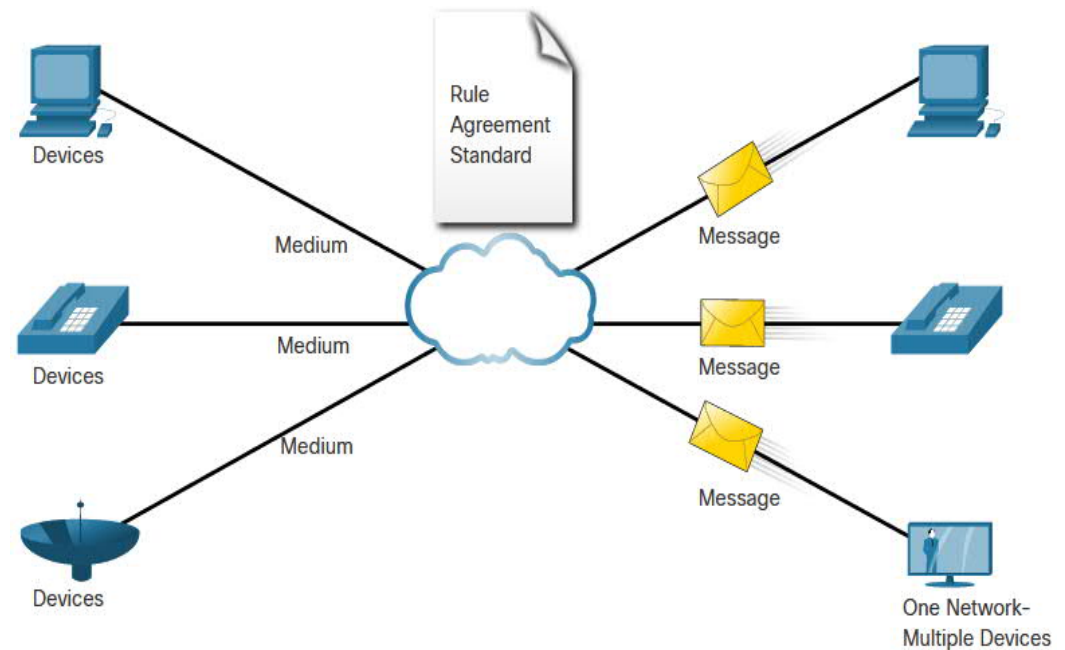
- Before converged networks, an organization would have been separately cabled for telephone, video, and data. Each of these networks would use different technologies to carry the signal.
- Each of these technologies would use a different set of rules and standards.



Internet Connections

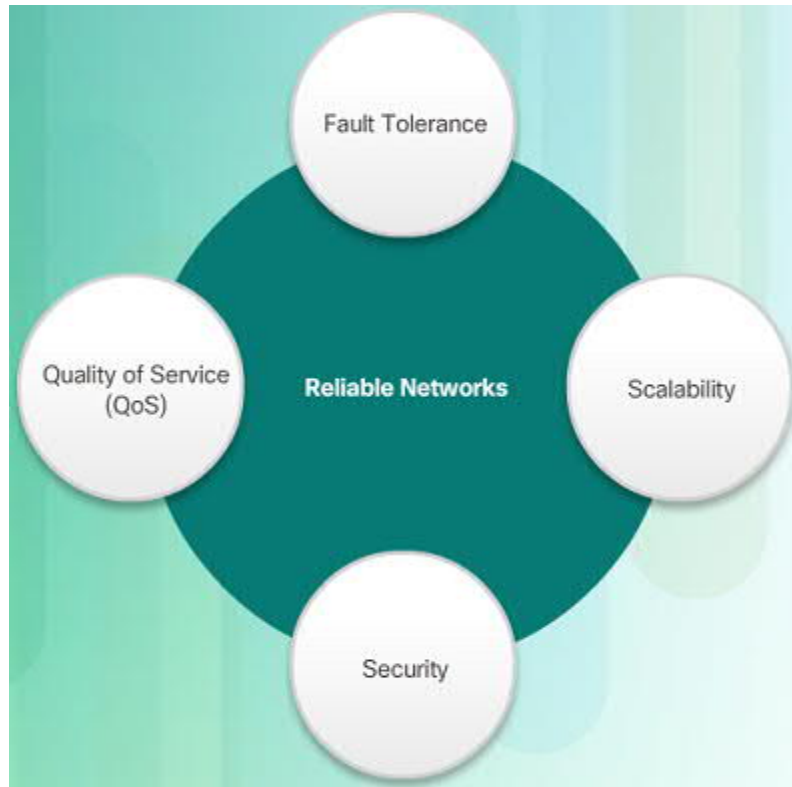
The Converging Network (Cont.)

- Converged data networks carry multiple services on one link including:
 - data
 - voice
 - video
- Converged networks can deliver data, voice, and video over the same network infrastructure.
- The network infrastructure uses the same set of rules and standards.



1.6 Reliable Networks

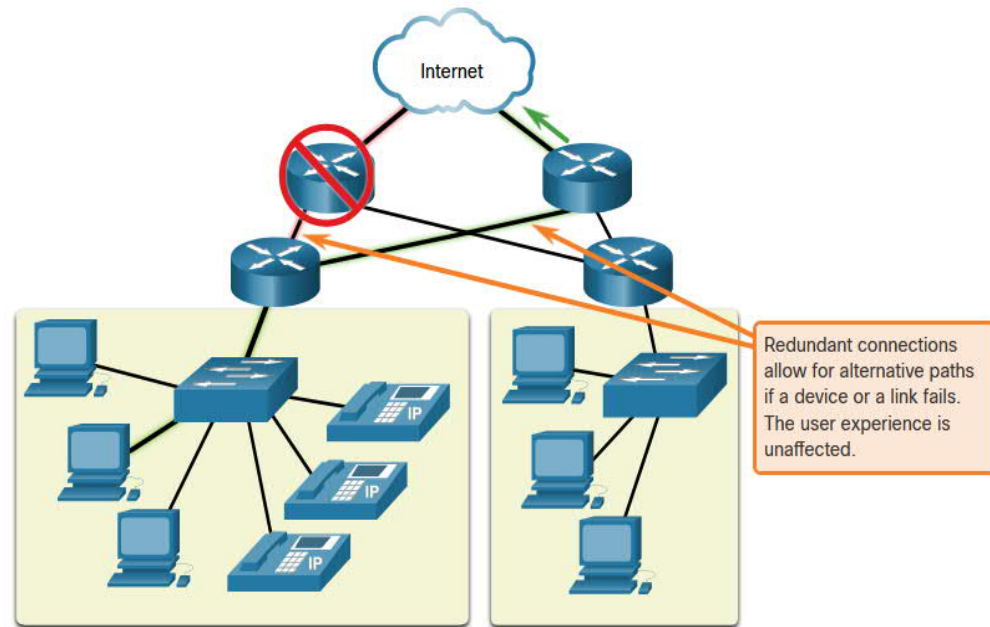
Reliable Network Network Architecture



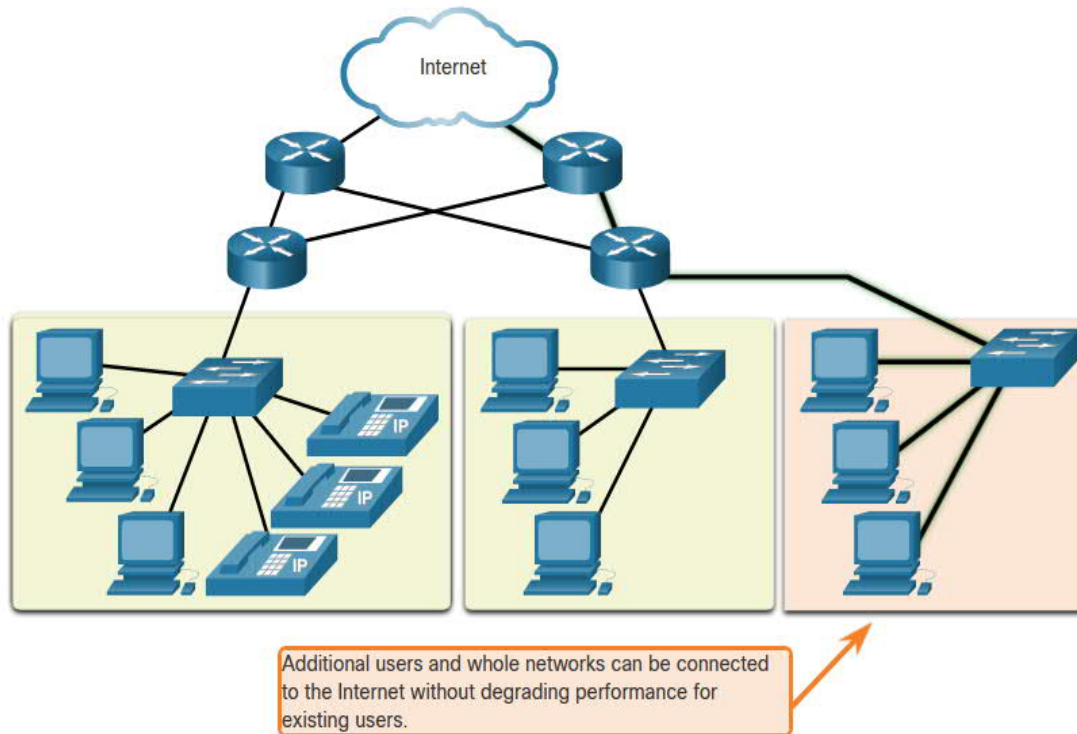
- Network Architecture refers to the technologies that support the infrastructure that moves data across the network.
- There are four basic characteristics that the underlying architectures need to address to meet user expectations:
 - Fault Tolerance
 - Scalability
 - Quality of Service (QoS)
 - Security

Reliable Network Fault Tolerance

- A fault tolerant network **limits the impact of a failure by limiting the number of affected devices.**
- Multiple paths are required for fault tolerance.
- **Reliable networks provide redundancy** by implementing a packet switched network:
 - Packet switching splits traffic into packets that are routed over a network.
 - Each packet could theoretically take a different path to the destination.
- This is not possible with circuit-switched networks which establish dedicated circuits.



Reliable Network Scalability

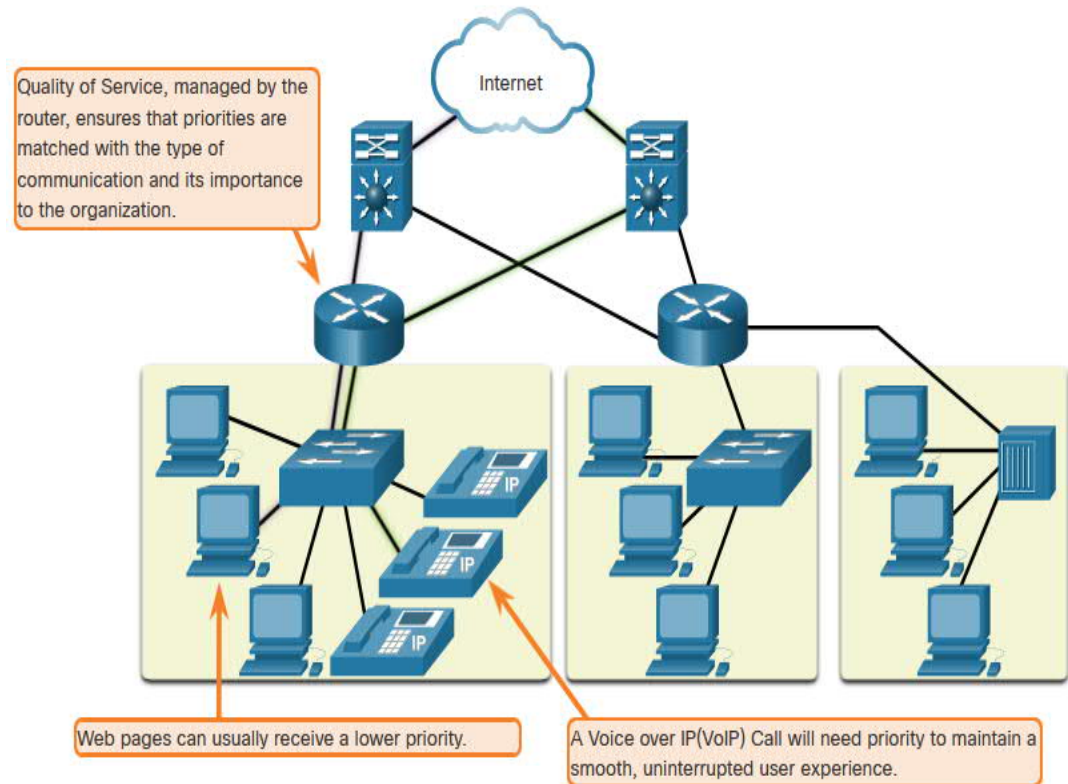


A scalable network can expand quickly and easily to support new users and applications without impacting the performance of services to existing users.

Network designers follow accepted standards and protocols in order to make the networks scalable.

Reliable Network Quality of Service

- Voice and live video transmissions require higher expectations for those services being delivered.
- Have you ever watched a live video with constant breaks and pauses? This is caused when there is a higher demand for bandwidth than available – and QoS isn't configured.
- Quality of Service (QoS) is the primary mechanism used to ensure reliable delivery of content for all users.
- With a QoS policy in place, the router can more easily manage the flow of data and voice traffic.



Network Security

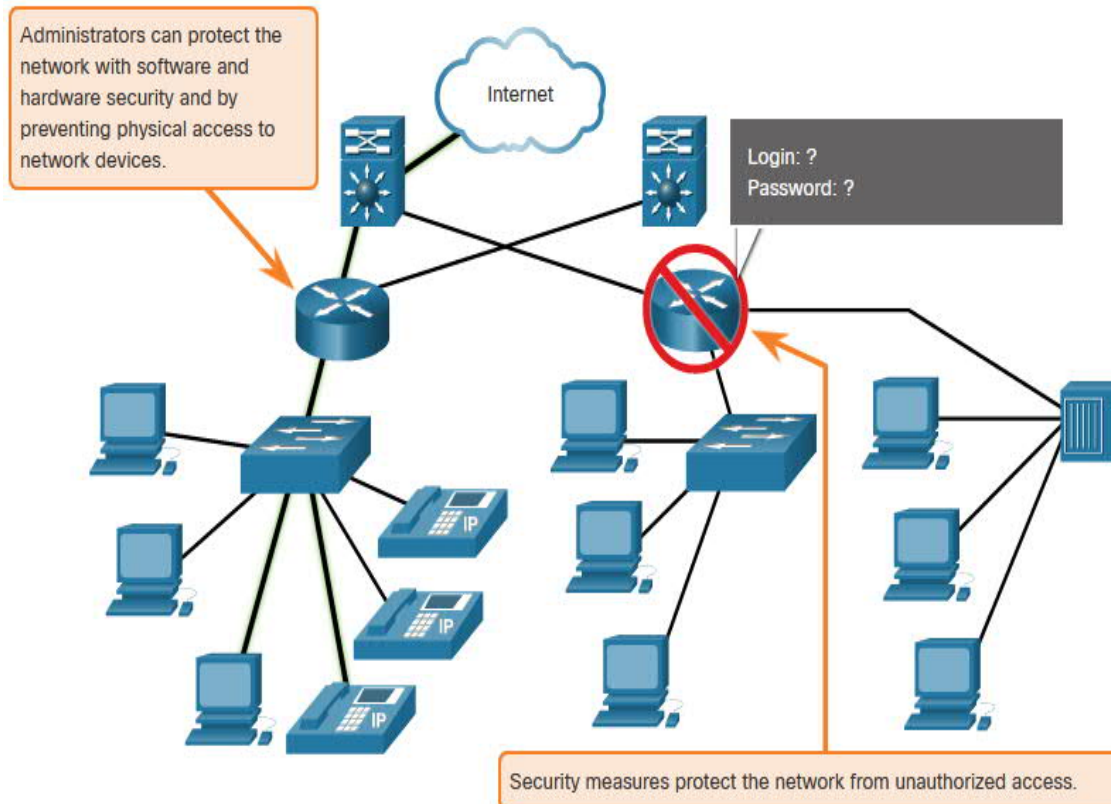
There are two main types of network security that must be addressed:

- **Network infrastructure security**
 - Physical security of network devices
 - Preventing unauthorized access to the devices
- **Information Security**
 - Protection of the information or data transmitted over the network

Three goals of network security:

- **Confidentiality** – only intended recipients can read the data
- **Integrity** – assurance that the data has not be altered with during transmission
- **Availability** – assurance of timely and reliable access to data for authorized users

Reliable Network Network Security



1.7 Network Trends

Network Trends

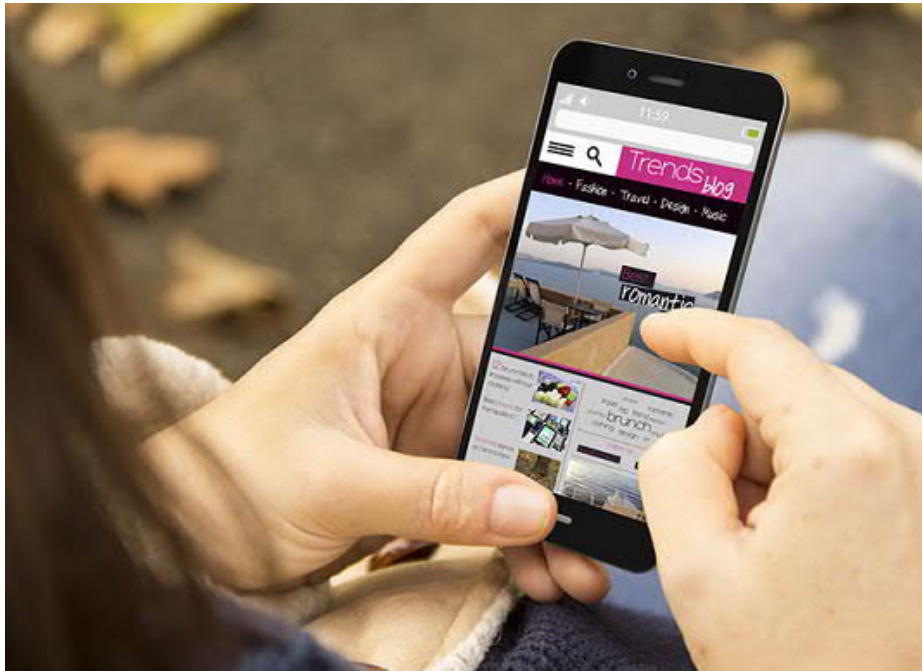
Recent Trends



- The role of the network must adjust and continually transform in order to be able to keep up with new technologies and end user devices as they constantly come to the market.
- Several new networking trends that effect organizations and consumers:
 - Bring Your Own Device (BYOD)
 - Online collaboration
 - Video communications
 - Cloud computing

Network Trends

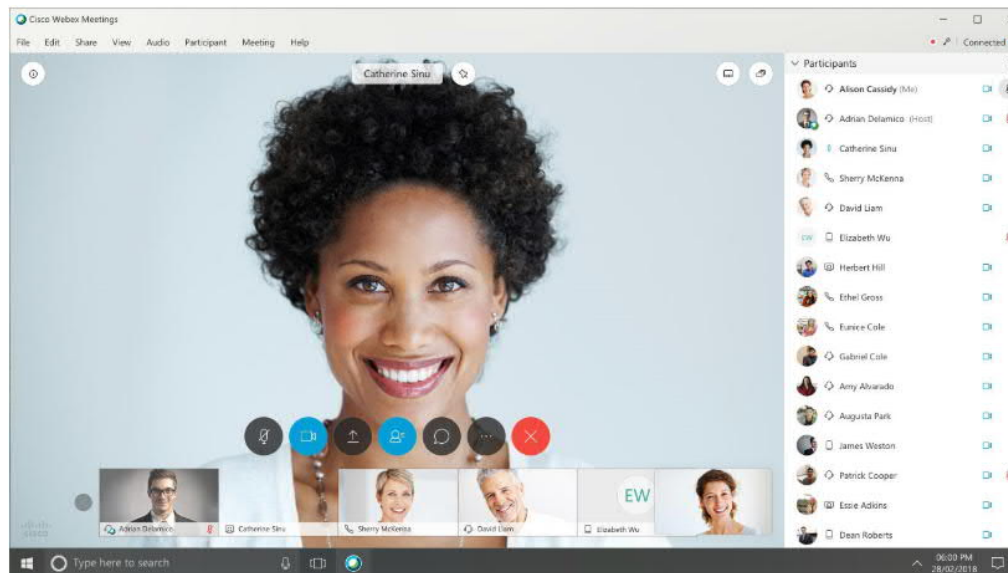
Bring Your Own Device



- Bring Your Own Device (BYOD) allows users to use their own devices giving them more opportunities and greater flexibility.
- BYOD allows end users to have the freedom to use personal tools to access information and communicate using their:
 - Laptops
 - Netbooks
 - Tablets
 - Smartphones
 - E-readers
- **BYOD means any device, with any ownership, used anywhere.**

Network Trends

Online Collaboration



- Collaborate and work with others over the network on joint projects.
- Collaboration tools including Cisco WebEx (shown in the figure) gives users a way to instantly connect and interact.
- Collaboration is a very high priority for businesses and in education.
- Cisco Webex Teams is a multifunctional collaboration tool.
 - send instant messages
 - post images
 - post videos and links

Network Trends

Video Communication

- Video calls are made to anyone, regardless of where they are located.
- Video conferencing is a powerful tool for communicating with others.
- Video is becoming a critical requirement for effective collaboration.
- Cisco TelePresence powers is one way of working where everyone, everywhere.



Network Trends

Cloud Computing

Cloud computing allows us to store personal files or backup our data on servers over the internet.

- Applications such as word processing and photo editing can also be accessed using the Cloud.
- Allows businesses to deliver to any device anywhere in the world.

Cloud computing is made possible by data centers.

- Smaller companies that can't afford their own data centers, lease server and storage services from larger data center organizations in the Cloud.

Network Trends

Cloud Computing (Cont.)

Four types of Clouds:

- **Public Clouds**
 - Available to the general public through a pay-per-use model or for free.
- **Private Clouds**
 - Intended for a specific organization or entity such as the government.
- **Hybrid Clouds**
 - Made up of two or more Cloud types – for example, part custom and part public.
 - Each part remains a distinctive object but both are connected using the same architecture.
- **Custom Clouds**
 - Built to meet the needs of a specific industry, such as healthcare or media.
 - Can be private or public.

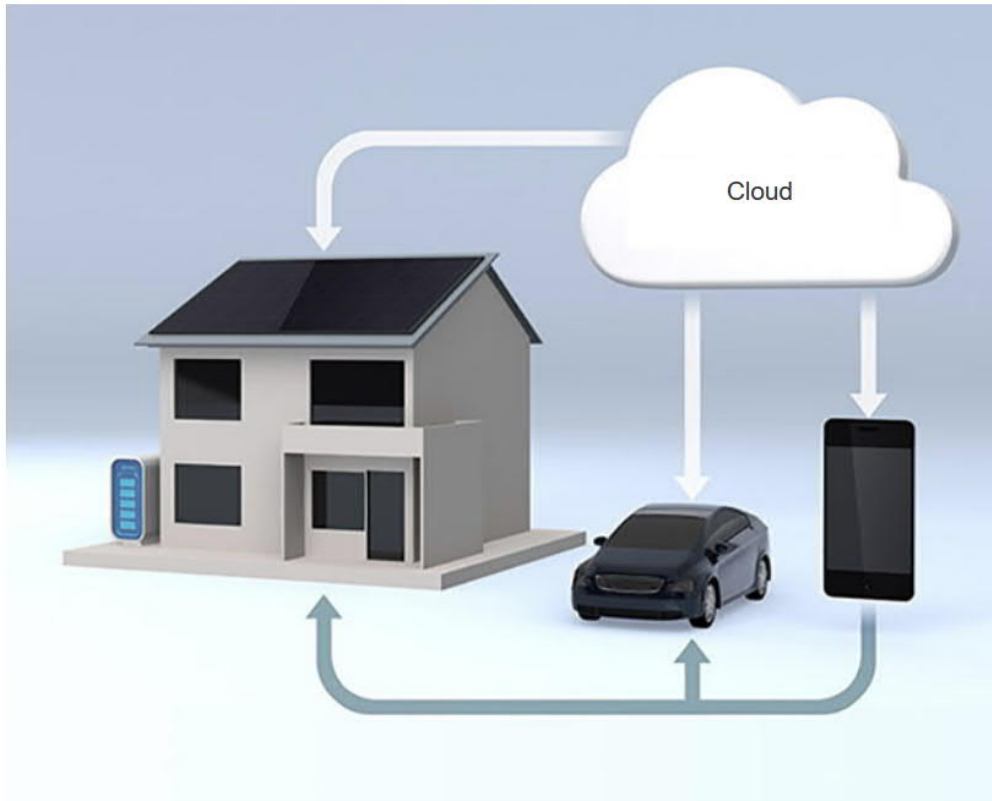
Network Trends

Cloud Computing (Cont.)



Network Trends

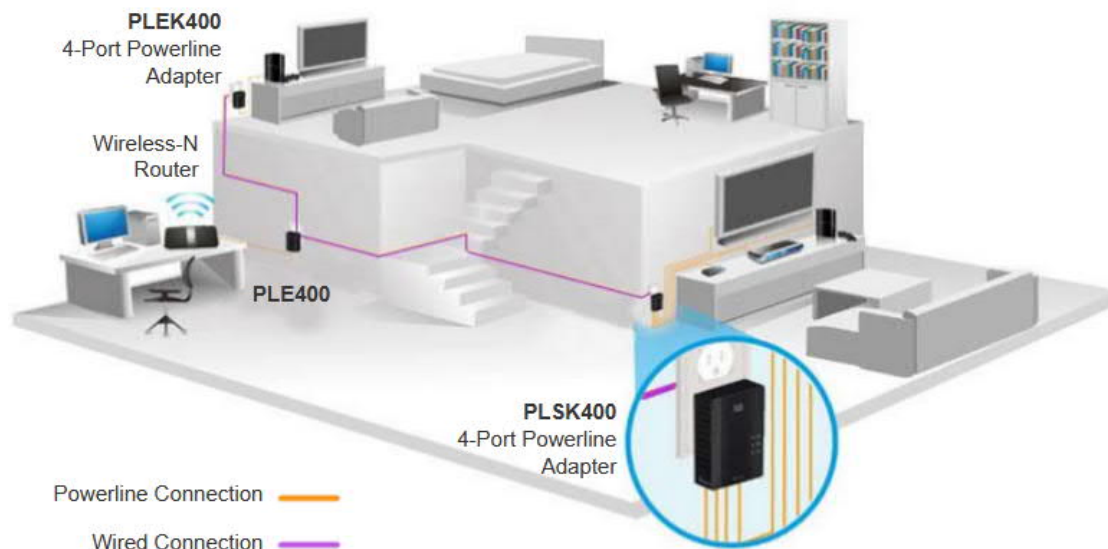
Technology Trends in the Home



- Smart home technology is a growing trend that allows technology to be integrated into every-day appliances which allows them to interconnect with other devices.
- Ovens might know what time to cook a meal for you by communicating with your calendar on what time you are scheduled to be home.
- Smart home technology is currently being developed for all rooms within a house.

Network Trends

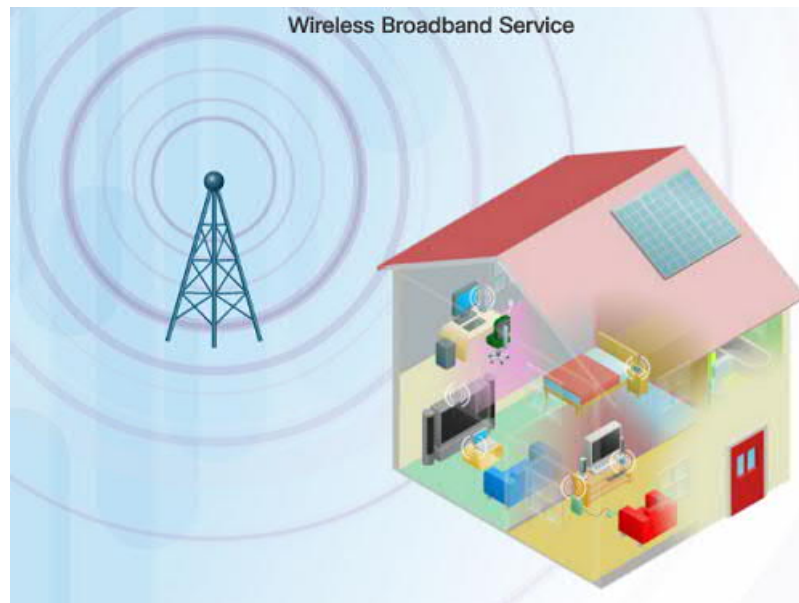
Powerline Networking



- Powerline networking can allow devices to connect to a LAN where data network cables or wireless communications are not a viable option.
- Using a standard powerline adapter, devices can connect to the LAN wherever there is an electrical outlet by sending data on certain frequencies.
- Powerline networking is especially useful when wireless access points cannot reach all the devices in the home.

Network Trends

Wireless Broadband



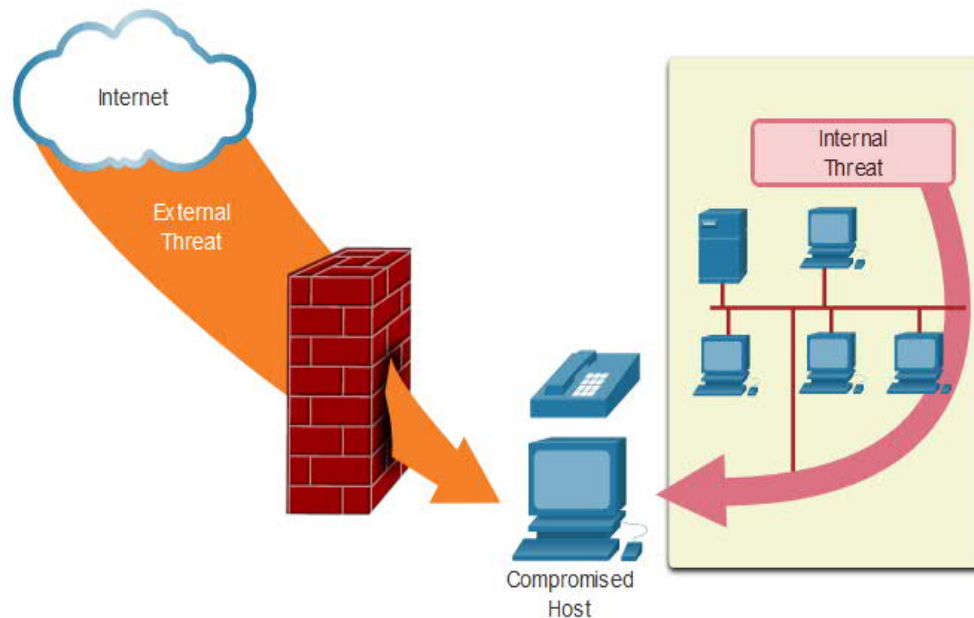
In addition to DSL and cable, wireless is another option used to connect homes and small businesses to the internet.

- More commonly found in **rural environments**, a Wireless Internet Service Provider (WISP) is an ISP that connects subscribers to designated access points or hotspots.
- Wireless broadband is another solution for the home and small businesses.
- Uses the same cellular technology used by a smart phone.
- An antenna is installed outside the house providing wireless or wired connectivity for devices in the home.

1.8 Network Security

Network Security

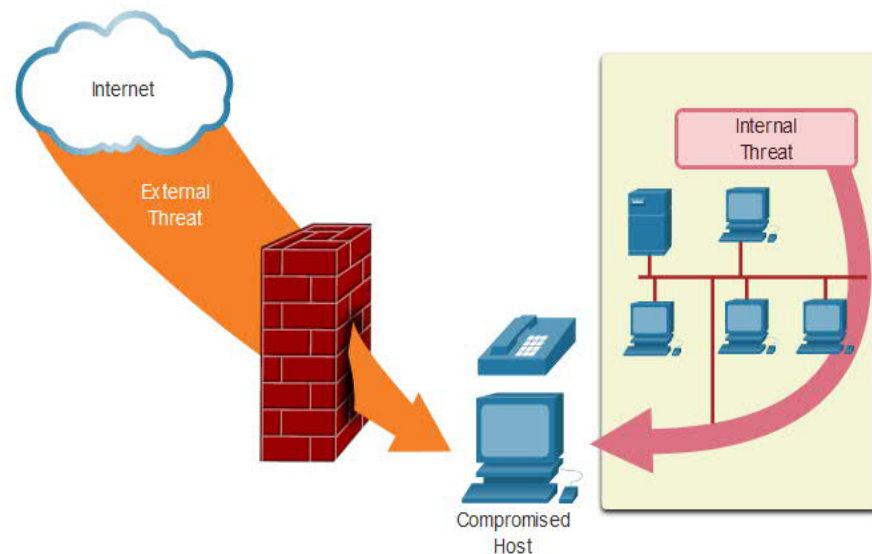
Security Threats



- Network security is an integral part of networking regardless of the size of the network.
- The network security that is implemented must take into account the environment while securing the data, but still allowing for quality of service that is expected of the network.
- Securing a network involves many protocols, technologies, devices, tools, and techniques in order to secure data and mitigate threats.
- Threat vectors might be external or internal.

Network Security

Security Threats (Cont.)



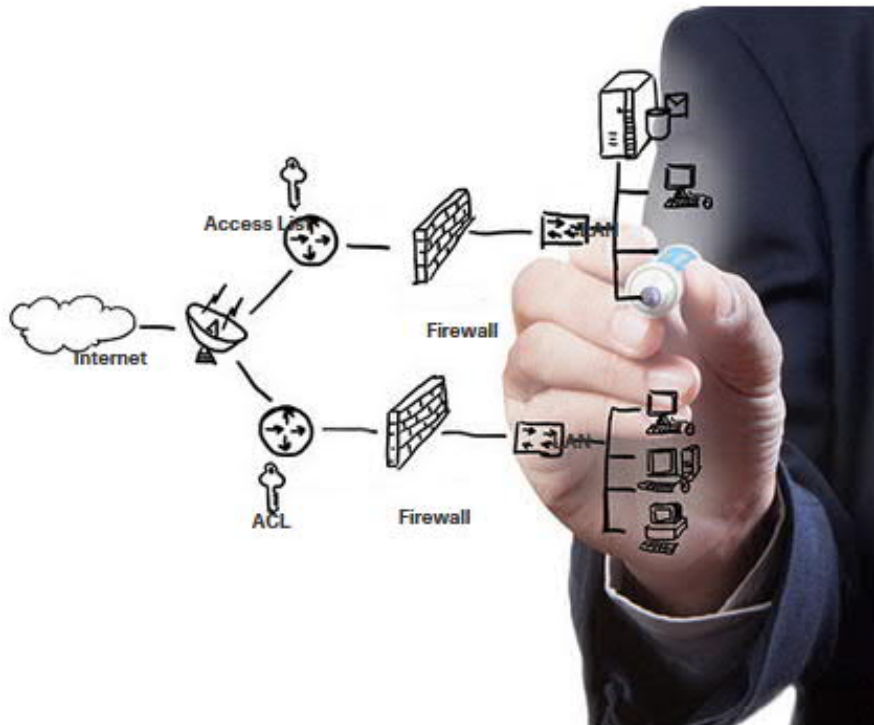
External Threats:

- Viruses, worms, and Trojan horses
- Spyware and adware
- Zero-day attacks
- Threat Actor attacks
- Denial of service attacks
- Data interception and theft
- Identity theft

Internal Threats:

- lost or stolen devices
- accidental misuse by employees
- malicious employees
- the internal users of the network cause the most security breaches.

Network Security Security Solutions

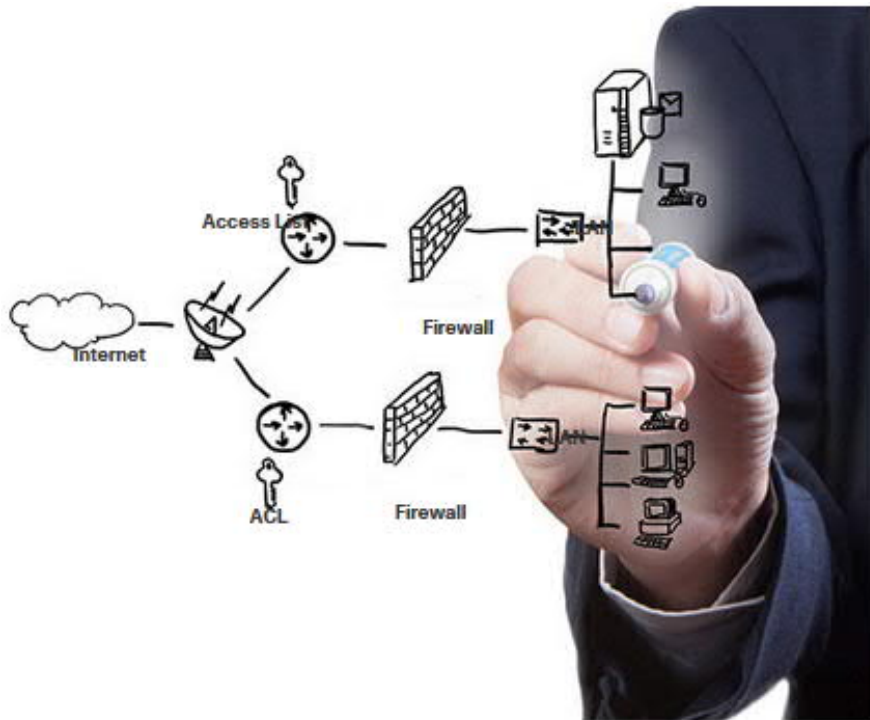


Security must be implemented in multiple layers using more than one security solution.

Network security components for home or small office network:

- **Antivirus** and antispyware software should be installed on end devices.
- **Firewall** filtering used to block unauthorized access to the network.

Network Security Security Solutions (Cont.)



Larger networks have additional security requirements:

- **Dedicated firewall** system to provide more advanced firewall capabilities.
- **Access control lists** (ACL) – used to further filter access and traffic forwarding
- **Intrusion prevention systems** (IPS) – used to identify fast-spreading threats such as zero-day attacks.
- **Virtual private networks** (VPN) – used to provide secure access for remote workers.

The study of network security starts with a clear understanding of the underlying switching and routing infrastructure.

Module Practice and Quiz

What did I learn in this module?

- Through the use of networks, we are connected like never before.
- All computers that are connected to a network and participate directly in network communication are classified as hosts.
- Diagrams of networks often use symbols to represent the different devices and connections that make up a network.
- A diagram provides an easy way to understand how devices connect in a large network.
- The two types of network infrastructures are Local Area Networks (LANs), and Wide Area Networks (WANs).
- SOHO internet connections include cable, DSL, Cellular, Satellite, and Dial-up telephone.
- Business internet connections include Dedicated Leased Line, Metro Ethernet, Business DSL, and Satellite.

What did I learn in this module? (Cont.)

- Network architecture refers to the technologies that support the infrastructure and the programmed services and rules, or protocols, that move data across the network.
- There are four basic characteristics of network architecture: Fault Tolerance, Scalability, Quality of Service (QoS), and Security.
- Recent networking trends that affect organizations and consumers: Bring Your Own Device (BYOD), online collaboration, video communications, and cloud computing.
- There are several common external and internal threats to networks.
- Larger networks and corporate networks use antivirus, antispyware, and firewall filtering, but they also have other security requirements: Dedicated firewall systems, Access control lists (ACL), Intrusion prevention systems (IPS), and Virtual private networks (VPN)
- The Cisco Certified Network Associate (CCNA) certification demonstrates your knowledge of foundational technologies.

Module 1

New Terms and Commands

- | | | |
|--|--|---|
| <ul style="list-style-type: none">• Peer-to-Peer File Sharing• Small Office/Home Office or SOHO• Medium to large network• Server• Client• Peer-to-Peer network• End device• Intermediary device• Medium• Network Interface Card (NIC)• Physical Port• Interface• Physical topology diagram | <ul style="list-style-type: none">• Logical topology diagram• Local Area Network (LAN)• Wide Area Network (WAN)• Internet• Intranet• Extranet• Internet Service Provider (ISP)• Converged networks• Network architecture• Fault tolerant network• Packet-switched network• Circuit-switched network• Scalable network• Quality of Service (Qos) | <ul style="list-style-type: none">• Network bandwidth• Bring Your Own Device (BYOD)• Collaboration• Cloud computing• Private clouds• Hybrid clouds• Public clouds• Custom clouds• Data center• Smart home technology• Powerline networking• Wireless Internet Service Provider (WISP)• Network architecture |
|--|--|---|

